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Degree: B.Tech., Semester: III-Course work
MID-SEMESTER EXAMINATION (ODD), September 2026
Course Title: Applied Linear Algebra
Course Code: CMMTC303

Duration: 1:30 Hours Max. Marks: 20

Note: Attempt all questions in given order only. Missing data/information (if any), maybe suitably assumed and mentioned in the answer.

Q. No.	Questions	Marks	CO
1a	Check whether the set $G = \left\{ \begin{pmatrix} 1 & \lambda \\ 0 & 1 \end{pmatrix} : \lambda \in \mathbb{R} \right\}$ under usual multiplication of matrices forms a group or not.	2	CO1
1b	Consider the ring $\mathbb{Z}_8$. (*i*) Check whether $\mathbb{Z}_8$ has zero divisors. (*ii*) Is $\mathbb{Z}_8$ a field? Justify.	2	CO1
2a	Show that the set $V = \{A \in M_2(\mathbb{R}) : \text{trace}(A) = 0\}$ forms a subspace of vector space $M_2(\mathbb{R})$.	2	CO1
2b	Suppose U and W are subspaces of a vector space V over the field F such that $\dim U = 4$, $\dim W = 5$, and $\dim V = 7$. Find the possible dimensions of $U \cap W$.	2	CO1
3a	For what real value(s) of k, the set of vectors $\{(k,1,1),\ (0,1,1,),\ (k,1,k)\}$ form a basis of $\mathbb{R}^3$?	2	CO1
3b	Find the range and kernel of the linear transformation T, where $T : P_2(\mathbb{R}) \longrightarrow P_3(\mathbb{R})$ is given by $T(p(x)) = xp(x)$ for all $p(x) \in P_2(\mathbb{R})$.	2	CO2
4a	Verify rank-nullity theorem for the linear transformation $T : \mathbb{R}^2 \to \mathbb{R}^4$ defined by $T(x,y) = (x+y,\ x-y,\ 2x,\ 2x-3y)$.	2	CO2
4b	Let $T : M_2(\mathbb{R}) \to M_2(\mathbb{R})$ be the linear transformation defined by $T\begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} a-b & b \\ c & 2c-d \end{pmatrix}$. Find a basis for the Range(T).	2	CO2
5a	Consider a linear transformation $T : P_2(\mathbb{R}) \to P_3(\mathbb{R})$ defined by $T(p(x)) = p'(x) + \int_0^x p(t)dt$. Determine whether T is invertible or not with proper justification.	2	CO2
5b	Find the matrix representation of a linear transformation $T : \mathbb{R}^3 \to P_3(\mathbb{R})$ defined by $T(a,b,c) = (a+2b) + bx + (c+a)x^2 + (a-b)x^3$ with respect to the standard bases of $\mathbb{R}^3$ and $P_3(\mathbb{R})$.	2	CO2